

Practical: 02

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```
#include<iostream>
#include<stdlib.h>
#include<omp.h>
using namespace std;
```

```
void bubble(int *, int);
void swap(int &, int &);
```

```
void bubble(int *a, int n)
{
    for(int i=0; i<n; i++)
    {
        int first = i % 2;

        #pragma omp parallel for shared(a,first)
        for(int j=first; j<n-1; j+=2)
        {
            if(a[j]>a[j+1])
            {
                swap(a[j], a[j+1]);
            }
        }
    }
}
```

```
void swap(int &a, int &b)
{
    int test;
    test=a;
    a=b;
    b=test;
}
```

```
int main()
{
    int *a,n;
    cout<<"\n Enter total no of elements=>";
    cin>>n;
    a=new int[n];
    cout<<"\n Enter elements=>";
    for(int i=0;i<n;i++)
    {
        cin>>a[i];
    }

    bubble(a,n);

    cout<<"\n Sorted array is=>";
```

```
    for(int i=0;i<n;i++)
    {
        cout<<a[i]<<endl;
    }

return 0;
}
```

OUTPUT:

```
vedant@LAPTOP-2AJF2JRG:~/Downloads$ g++ HPC2_1.cpp
vedant@LAPTOP-2AJF2JRG:~/Downloads$ ./a.out
```

Enter total no of elements=>5

Enter elements=>3

4
2
7
8

Sorted array is=>2

3
4
7
8

```

#include<iostream>
#include<stdlib.h>
#include<omp.h>
using namespace std;

void mergesort(int a[],int i,int j);
void merge(int a[],int i1,int j1,int i2,int j2);

void mergesort(int a[],int i,int j)
{
    int mid;
    if(i<j)
    {
        mid=(i+j)/2;

        #pragma omp parallel sections
        {

            #pragma omp section
            {
                mergesort(a,i,mid);
            }

            #pragma omp section
            {
                mergesort(a,mid+1,j);
            }
        }
        merge(a,i,mid,mid+1,j);
    }
}

void merge(int a[],int i1,int j1,int i2,int j2)
{
    int temp[1000];
    int i,j,k;
    i=i1;
    j=i2;
    k=0;
    while(i<=j1 && j<=j2)
    {
        if(a[i]<a[j])
        {
            temp[k++]=a[i++];
        }
        else
        {
            temp[k++]=a[j++];
        }
    }

    while(i<=j1)
    {
        temp[k++]=a[i++];
    }
}

```

```

        while(j<=j2)
        {
            temp[k++]=a[j++];
        }

        for(i=i1,j=0;i<=j2;i++,j++)
        {
            a[i]=temp[j];
        }
    }

int main()
{
    int *a,n,i;
    cout<<"\n enter total no of elements=>";
    cin>>n;
    a= new int[n];

    cout<<"\n enter elements=>";
    for(i=0;i<n;i++)
    {
        cin>>a[i];
    }

    mergesort(a, 0, n-1);

    cout<<"\n sorted array is=>";
    for(i=0;i<n;i++)
    {
        cout<<"\n"<<a[i];
    }

    return 0;
}

```

OUTPUT:

```

vedant@LAPTOP-2AJF2JRG:~/Downloads$ g++ HPC2_2.cpp
vedant@LAPTOP-2AJF2JRG:~/Downloads$ ./a.out

```

enter total no of elements=>6

enter elements=>4

9
8
2
6
4

sorted array is=>

2
4
4
6
8
9